Université Grenoble Alpes / Ensimag

Internship Report

Phage-Host Classification Using Conformal Prediction????

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Abstract

Acknowledgements

ENGLISH VERSION

Version française

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1 Introduction

The rapid rise of antibiotic-resistant bacteria is one of the major challenges that global medicine is facing today. To overcome this issue, researchers are going to alternative solutions, one of which is phage therapy. Phage therapy relies on using the naturally occurring viruses to target and destroy the harmful bacteria.

However, To make this therapy a viable alternative, we first need to understand the basic biology of these viruses. Bacteriophages, or phages are viruses that infect bacterial hosts that and are one of the most abundant entities on the planet. Each phage contains a collection of structural and functional proteins that collectively determine the bacterial strains that the phage can infect - this property is called the host range of the phage.

These phages are highly host-specific: a phage which infects one bacterial host may have very little to no effect on another. This particular property of theirs has naturally generated interest in modelling and applications in the context of....... . Therefore a significant problem in the phage biology is the prediction of the bacterial hosts associated with a given phage. Accurate host-predictions can have applications in .....

complete sentence (treatment???)

find applications

The host specificity of phages is one of phage therapy's greatest advantage, but at the same time, it also brings a very significant practical challenge. While it allows us to target specific harmful bacteria without destroying the beneficial ones, it also causes probelms in finding exactly the appropriate phage for a specific infection. Doing this by only experimental screening can be time-consuming and expensive. This motivates the development of reliable computational models that are capable of predicting these phage-host interactions directly from the protein level information.

The existing computational tools, however have a very important limitation: they typically only produce point predictions, which are oftern represented as a scalar confidence score between 0 and 1 with no associated measure of uncertainty. This lack of having an uncertainty quantification is problematic in clinical contexts, where failing to identify a true host could mean missing a life-saving treatment. Consequently, to build trust in automated host prediction, we need a mathematical framework that is capable of quantifying its own predictive uncertainty and provides reliable confidence guarantees.

The primary objective of this report is to address this limitation by developing an uncertainty-aware framework for phage-host prediction using **Conformal Prediction**. Conformal prediction provides a distribution-free framework that allows us to transform standard point predictions into prediction sets that are guaranteed to contain the true bacterial host with a user-defined confidence level.

In this project, we build our phage representations by aggregating collections of protein-level functional scores. This process naturally produced high-dimensional, structured score vectors, as different proteins and functions vary in their data availability. These score vectors also ofter contain missing or noisy information. To handle these challenges, we explored geometric conformal prediction methods that can construct reliable prediction sets even when dealing with incomplete biological data. As an intermediate step to validate our geometric approach in a controlled environment, we first evaluated the framework on a binary Gram-type classification task before moving on to the full multi-host prediction problem.

2 Background & State of the Art

2.1 Host Prediction Problem

Bacteriophages infect the bacterial hosts through a series of highlt

2.2 The Host Prediction Problem

2.3 State of the Art

2.4 Conformal Prediction

- framework
- coverage guarantee
- exchnagebility

2.5 The CSA Method

- ensemble score aggregation
- calibration
- envelope

2.6 Related Work???????

3 Methods

3.1 Problem Formalization

- Prediction sets - Scores - Envelope design

3.2 Envelope Contruction

3.2.1 CSA Calibration

3.2.2 Tau Computation

- one calculation, not binary search

3.3 Three envelope construction methods

- using conconvex envelopes

3.3.1 Radial Method

3.3.2 Strip Method

3.3.3 1D

3.4 $\hat{t}$ Calculation

- class conditional (why and how)

4 Data ....

4.1 Data Description

4.2 Protein to Phage Aggregation

- mean pooling - NaN handling

4.3 Train-Test Split

- handling the bias (currently working)

5 Experiments and Results

5.1 2D Synthetic Data

- correlated, uncorrelated etc predictors - K dimensions - multiple hosts

5.2 Gram Type Classification

using the three envelopes .... - Results: coverage by class, set size distribution, singleton accuracy - analysis of hard cases: ambiguous gram-positive phages

6 Conclusion and Future Work

6.1 Summary.....

6.2 Limitations

- gram-pos coverage -